CNENGR202 Instrumentation and Control

Part 2 – T1 Tutorial session

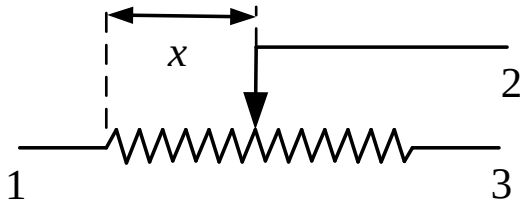
Prof. Zhang Liwei 张立伟 教授

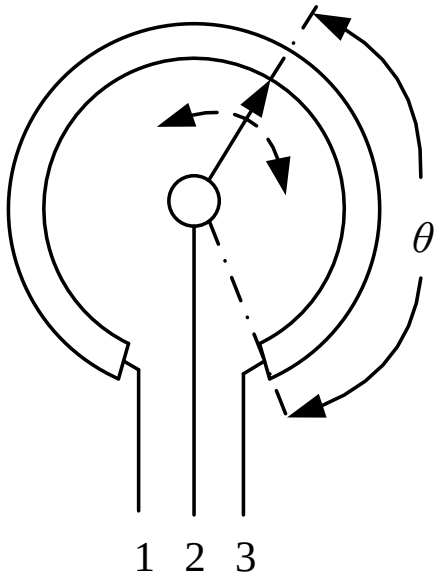
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Potentiometric sensors

- Two configurations: linear and rotary

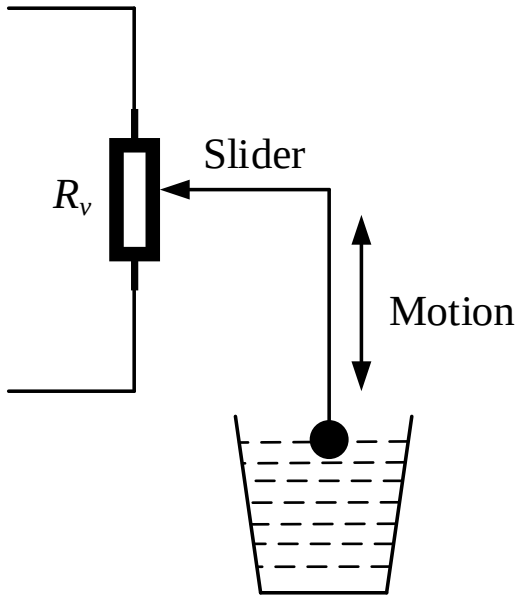




- Resistance change is the result of position variation (x or θ) of a movable contact (wiper) on a fixed resistor, resulting in an output voltage change

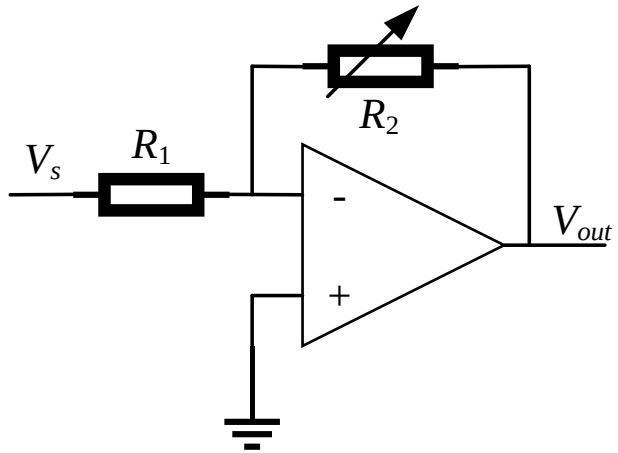
Potentiometric water level sensor

- A potentiometric sensor for water level measurement. If the resistance changes linearly from 0 to 2 kΩ over the entire water level range, (1) develop a measurement circuit to provide a linear, 0 to 10 output when the water level changes from 0 to 80mm; (2) if the sensor's output is 7.5V, what is the water level?



Solution

(1) The op-amp circuit as shown in right can be used for this measurement with R_1 = 2 kΩ (the maximum resistance of the sensor), V_S = 10 V, and R_2 being the potentiometer.



$$V_{out} = -\frac{R_2}{R_1}V_s = -\frac{R_2}{2000}\times 10 = -0.005R_2$$

We can ignore the sign

As R_2 varies from 0 to 2 kΩ, the output voltage V_{out} will change linearly from 0 to 10 V

Solution

(2) When V_{out} is 7.5 V, R_2 is

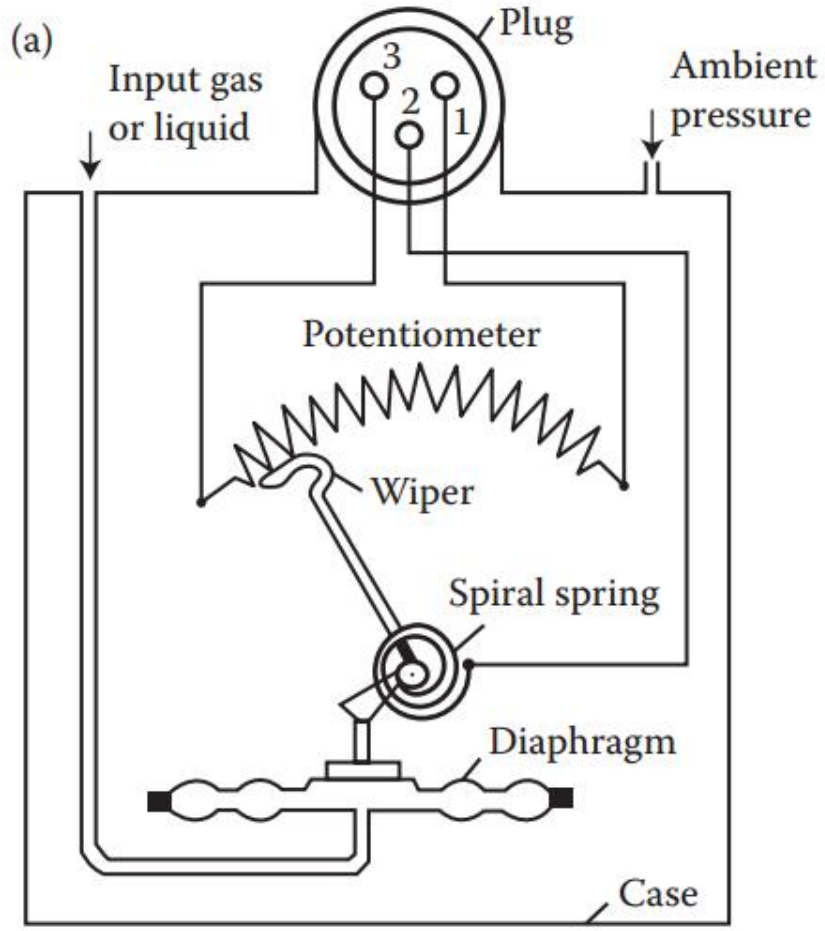
$$R_2 = \frac{V_{out}}{0.005} V_s = \frac{7.5}{0.005} = 1.5\text{k}\Omega$$

Since the water level x is proportional to the resistance change R_2, therefore

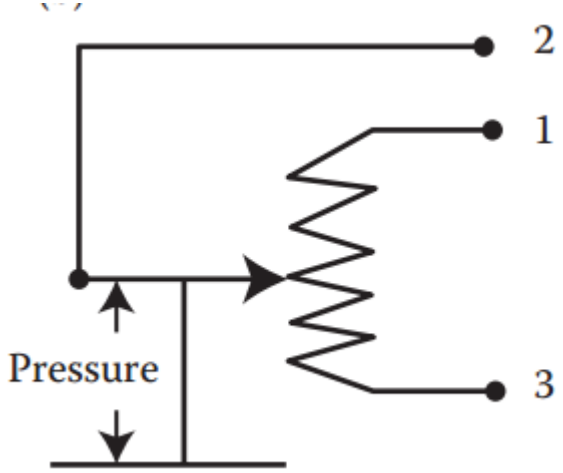
$$\frac{x - X_{\min}}{X_{\max} - X_{\min}} = \frac{R_2 - R_{\min}}{R_{\max} - R_{\min}} \Rightarrow \frac{x - 0}{80 - 0} = \frac{R_2 - 0}{2000 - 0}$$

$$x = 0.04R_2 = 0.04 \times 1500 = 60\text{mm}$$

Potentiometric Pressure Sensor

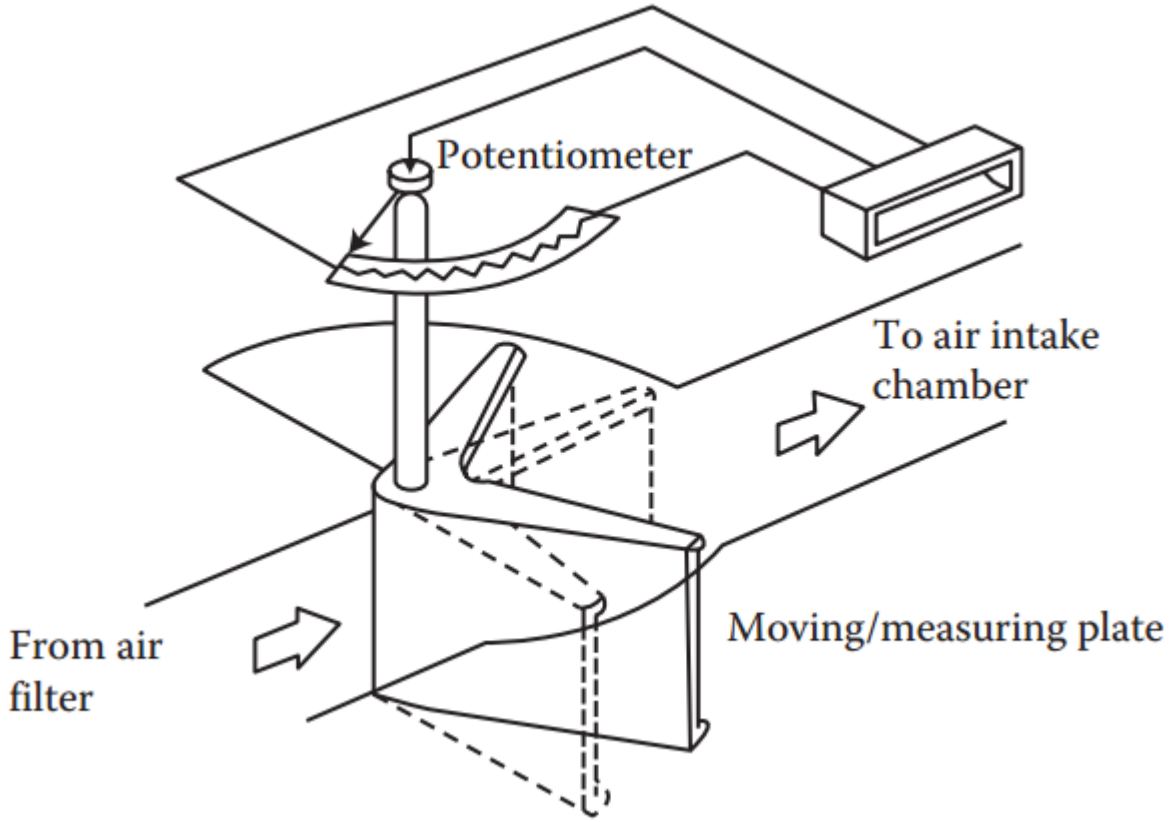


A rotary potentiometric pressure sensor



a linear potentiometric pressure sensor

Potentiometric Airflow Sensor



Example

Among RTDs, thermistors, and thermocouples, which temperature sensor(s) can

A. Measure a temperature that is below 0℃: _____

B. Measure a temperature that is above 1000℃: ____

C. Be made the smallest: _____

D. Have a direct voltage output (without using a voltage divider circuit): ____

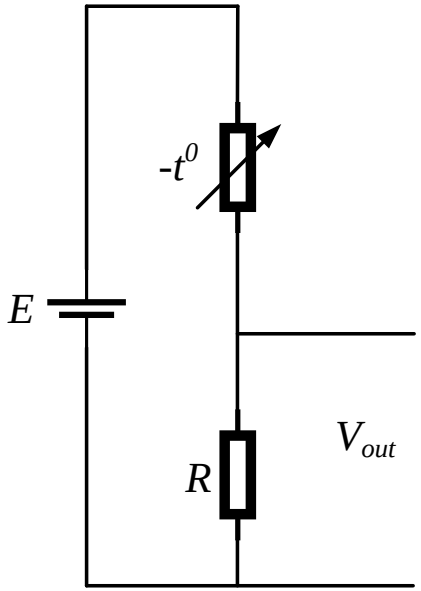
Example

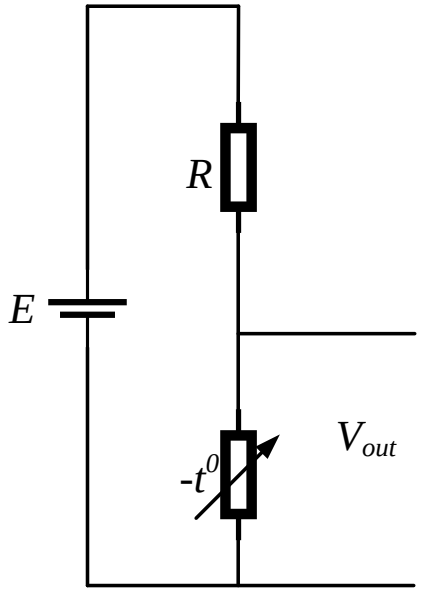
- Match the materials in the left column with sensors in the right column

Materials	**Sensors**
CdS	RTD
Pt100	Piezoresistive sensor
Ferry alloy	Photoresistive sensor

Example

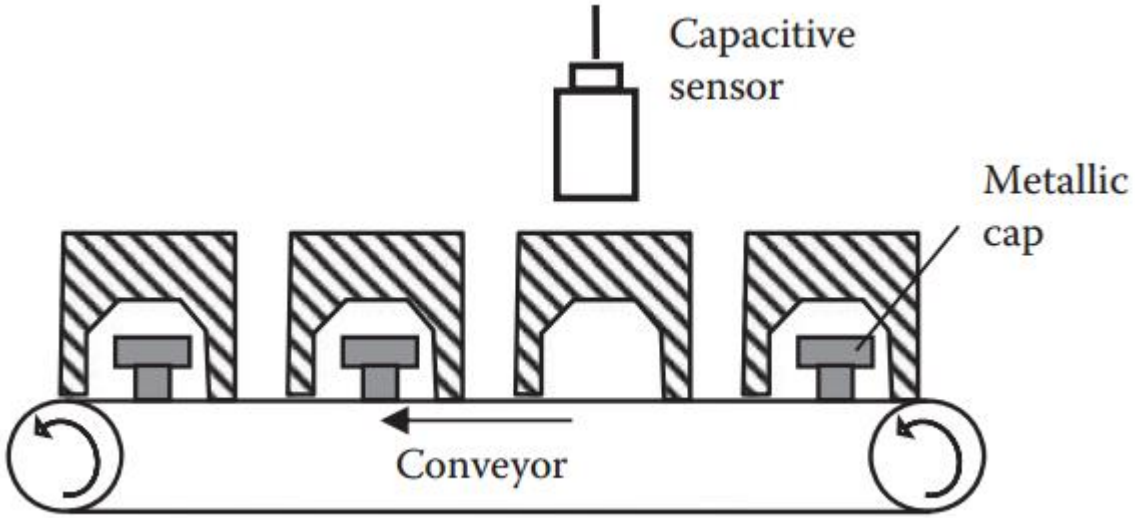
- A fire sensor circuit is designed to deliver a high voltage to trigger an alarm when a hot condition is detected. Which of the circuits should be chosen for this purpose? (Note: " $-t^0$" in the figure indicates an NTC sensor)





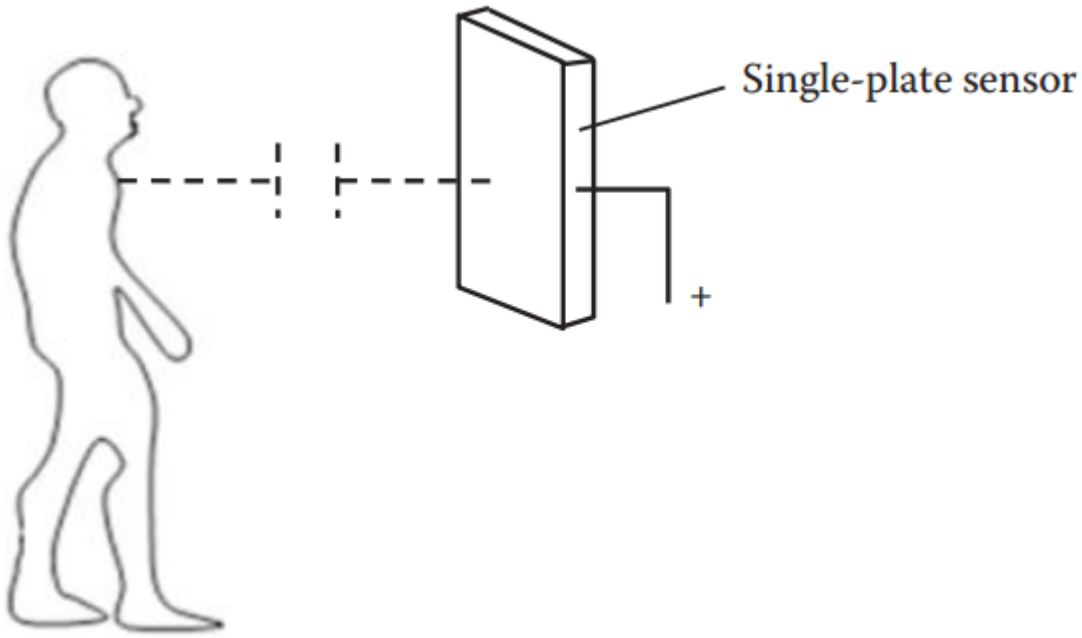
Example

- Detect the presence or absence of metallic parts in completed assemblies



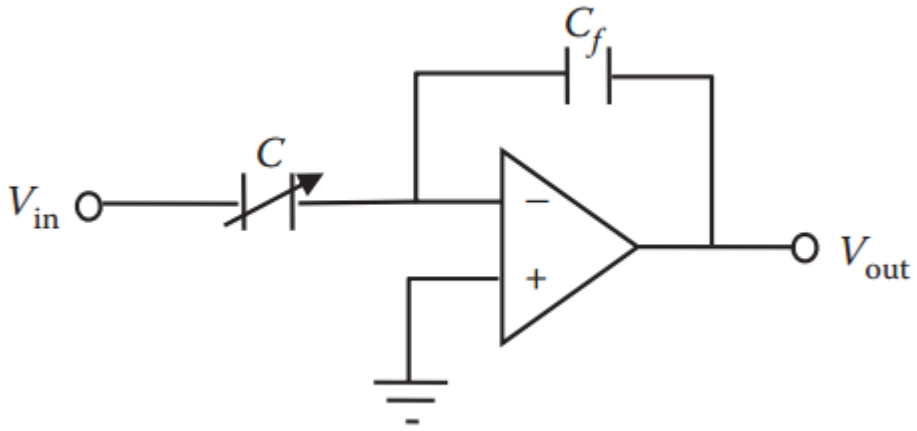
Example

- The capacitance increases when the human gets closer to the metal plate

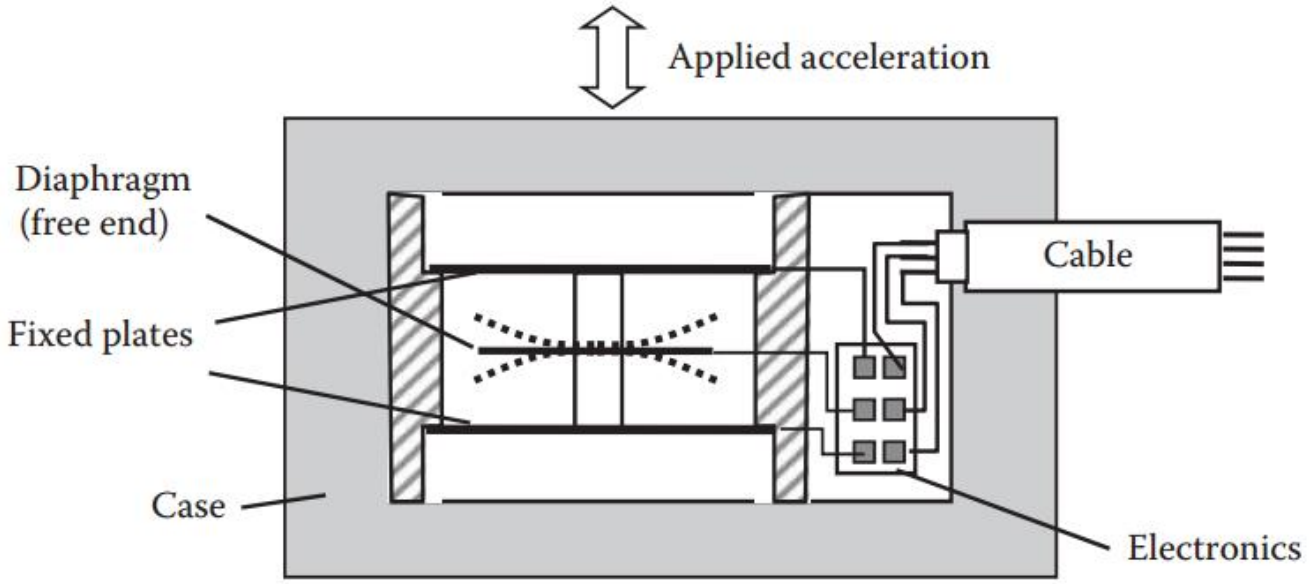


Capacitive Pressure Sensor

- A typical spacing-variation-based capacitive pressure sensor consists of a fixed plate and a flexible plate (diaphragm)



Capacitive Accelerometer -1



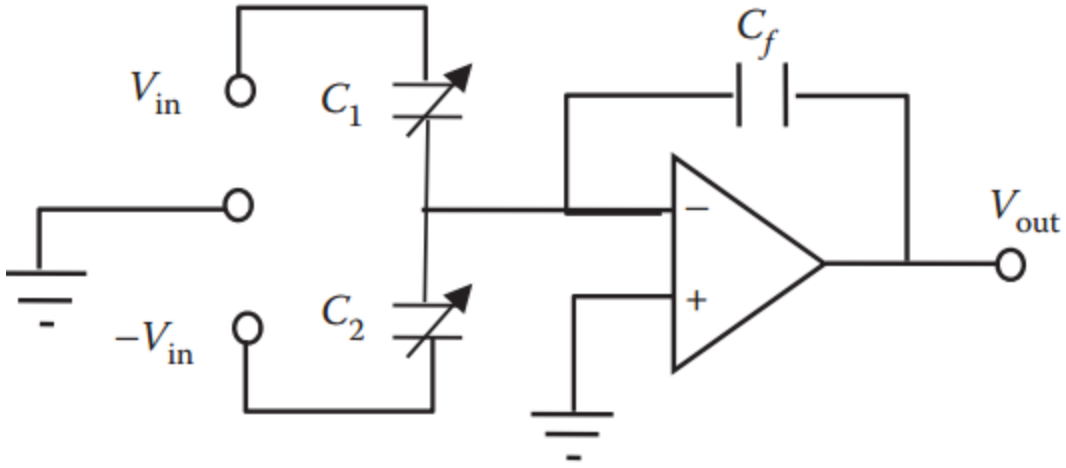
Capacitive Accelerometer -2

$$V_{out} = -\frac{C_1 - C_2}{C_f} V_{in} = \frac{2d_0 x}{d_0^2 - x^2} \frac{C_0}{C_f} V_{in}$$

$$C_0 = \frac{\varepsilon_0 \varepsilon_r S}{d_0}$$

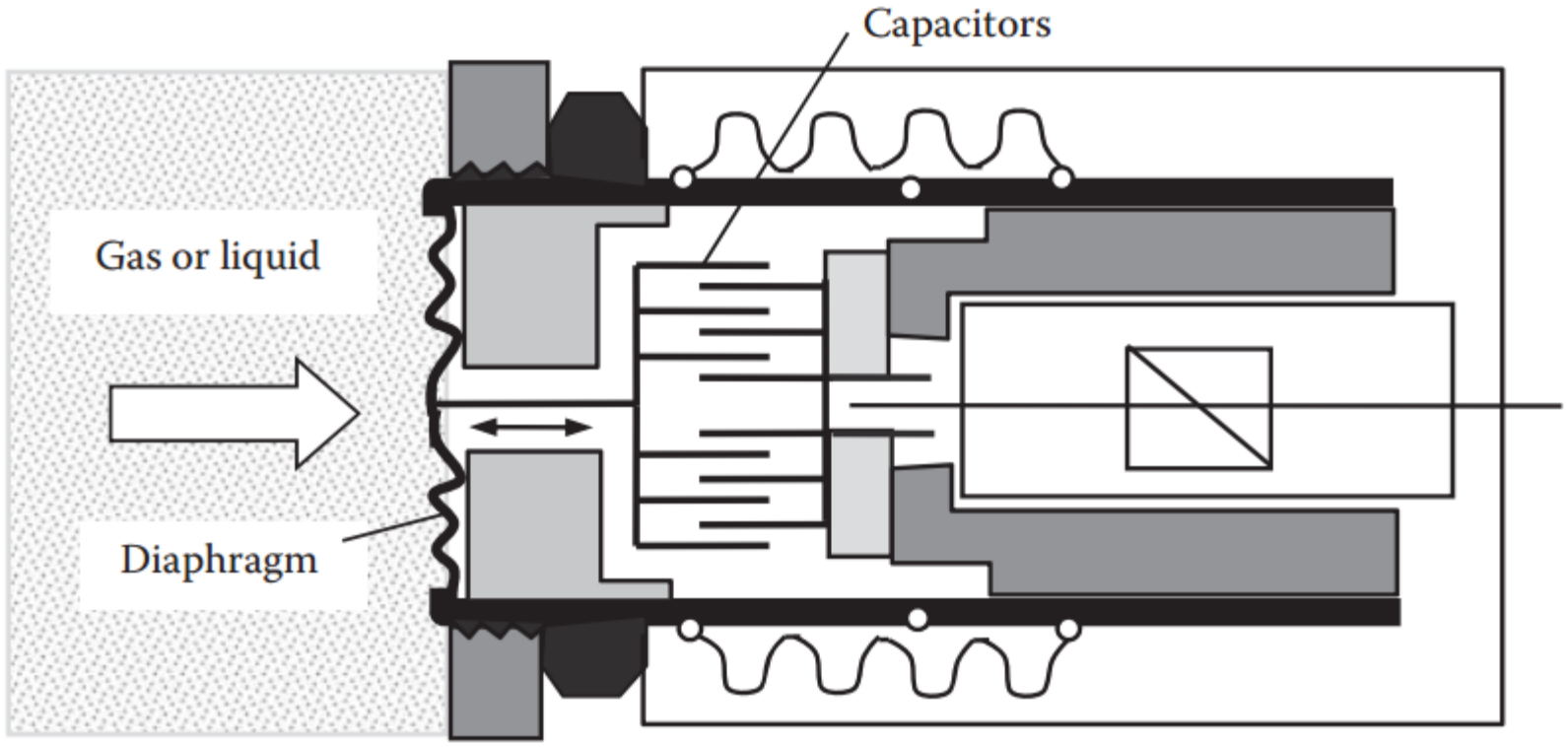
$$C_1 = \frac{\varepsilon_0 \varepsilon_r S}{d_0 + x}$$

$$C_2 = \frac{\varepsilon_0 \varepsilon_r S}{d_0 - x}$$



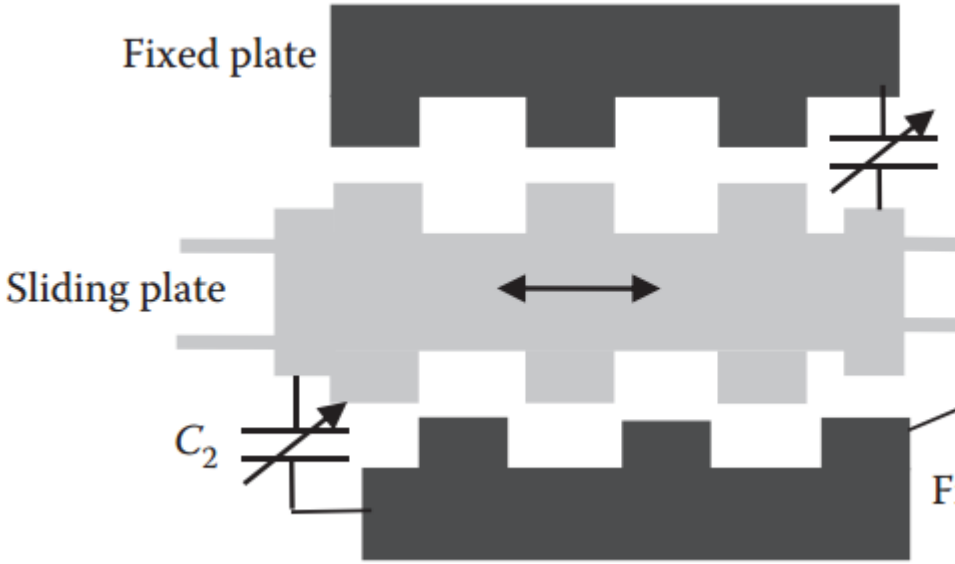
Capacitive Pressure sensor

- An area-variation-based capacitive pressure sensor
- It consists of multiple plates and is more robust than other types of pressure sensors



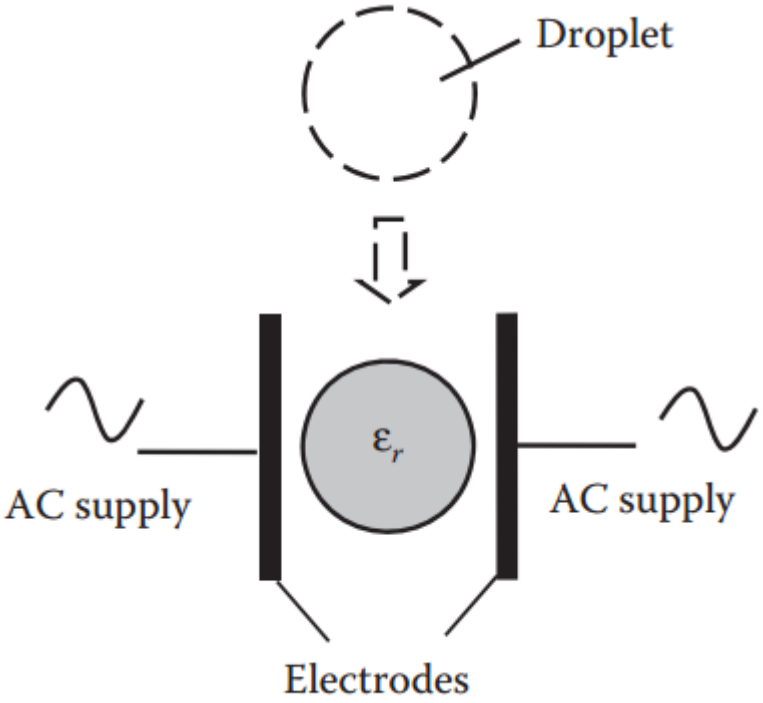
Capacitive Position sensor

- An area-varying capacitive position sensor
- It consists of two fixed parallel plates (the top and the bottom plates) and a sliding plate between them



Droplet Detector

- A free droplet is introduced between two plates of an open capacitor
- The presence of the droplet changes the dielectric property between the plates, causing a significant change in capacitance



Concept Problems -1

Capacitive sensors are designed based on the following variation(s) *except*_____

A. Space between the plates
B. Dielectric material property
C. Overlap area between plates
D. Excitation voltage

Concept Problems -2

Which of the following statements about a capacitor is true?

A. A capacitor is an energy storage element.
B. All capacitors are polarized.
C. Capacitor is an active component.
D. A capacitor works properly only under an AC source

Concept Problems -3

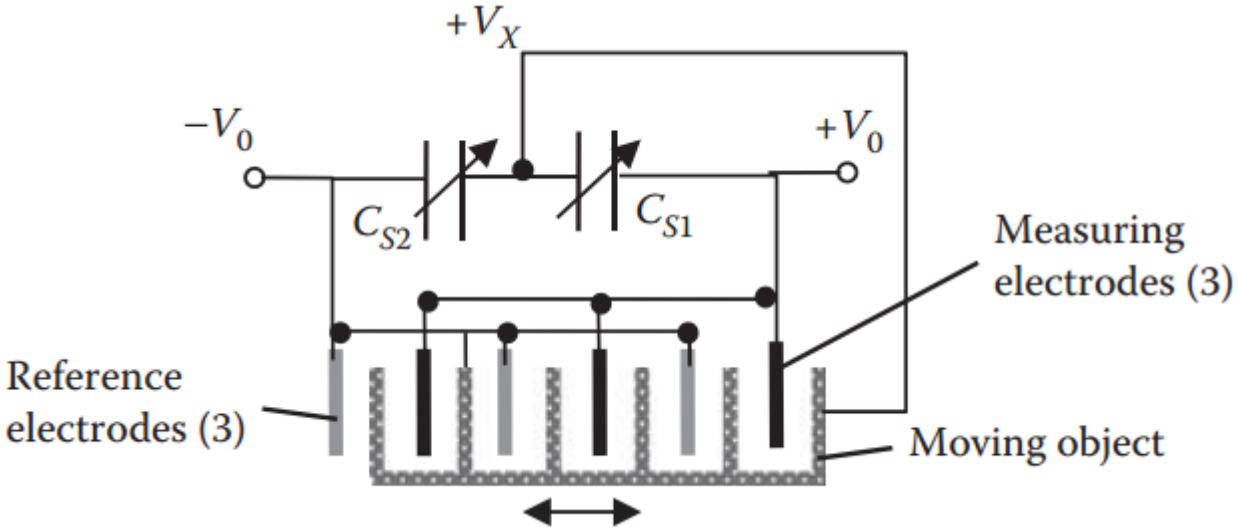
A multiplate capacitive sensor has a circuit diagram shown in the figure. If the object moves right, which of the following best describes the resultant capacitance C_{S1} (where C_0 is the capacitance of C_{S1} when the object is in the middle position)

A. $C_{S1} = C_0 + \Delta C$ and $\Delta C > 0$

B. $C_{S1} = C_0 - \Delta C$ and $\Delta C > 0$

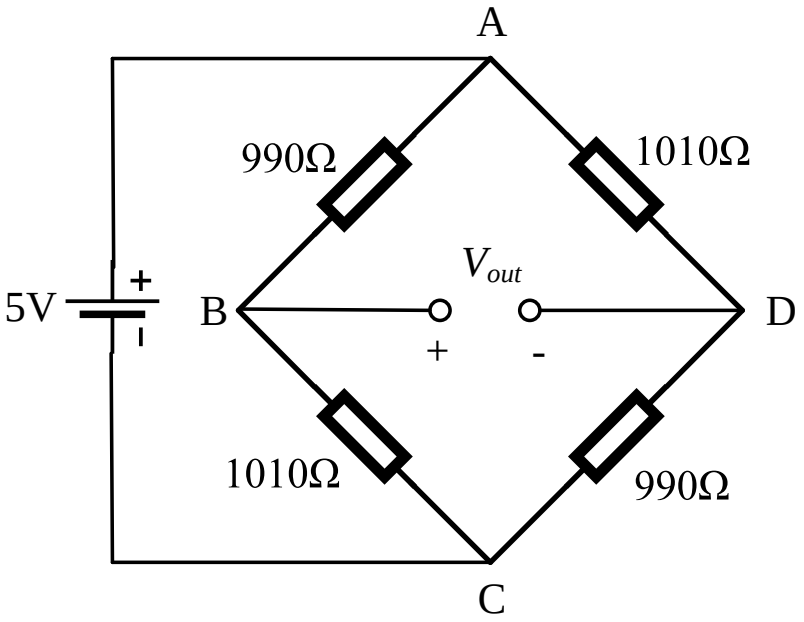
C. $C_{S1} = C_0 + \Delta C$ and $\Delta C < 0$

D. $C_{S1} = C_0 - \Delta C$ and $\Delta C < 0$

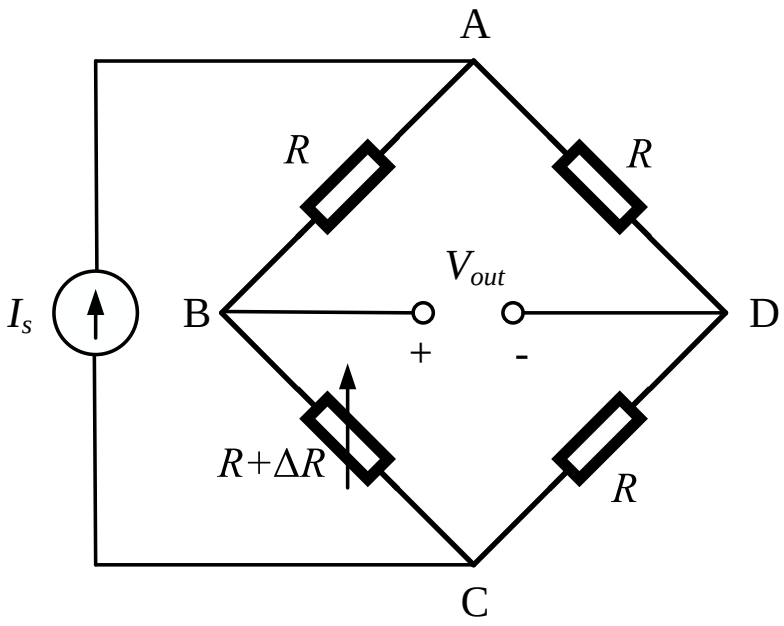


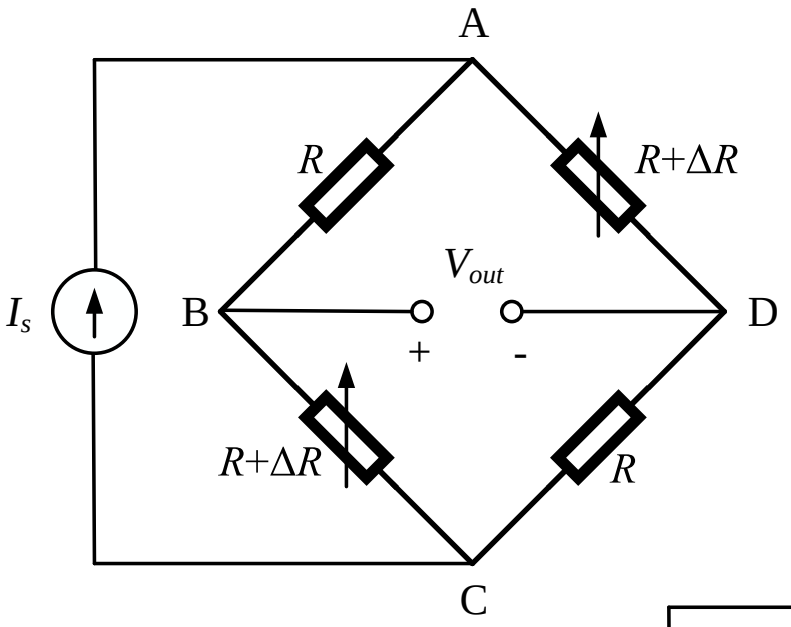
Example

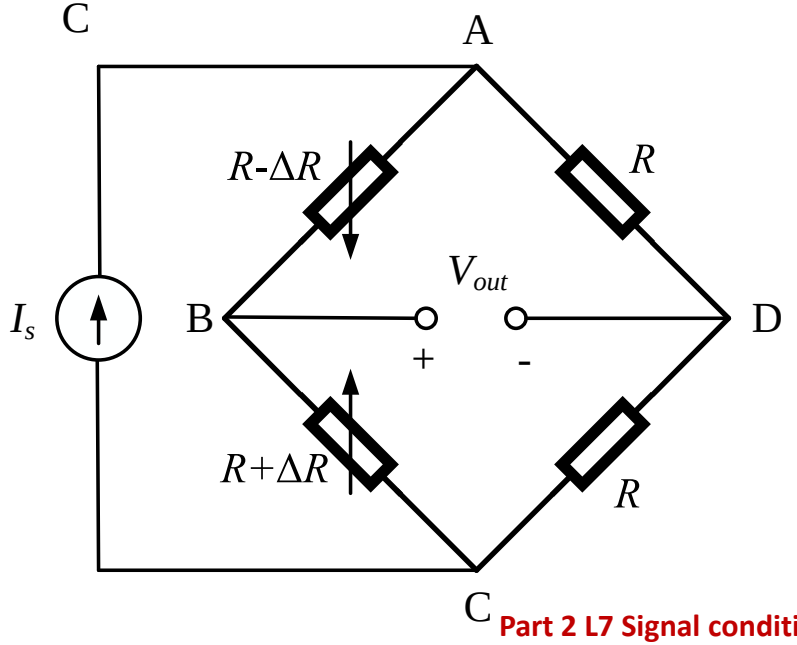
A resistive sensor has a Wheatstone bridge configuration. Its sensing element on each arm has an actual resistance value shown in the figure and a nominal resistance of 1 kΩ where $\Delta R = \pm 10\ \Omega$. If the sensor's sensitivity is 10 mV/V and the excitation voltage is 5 V, find the sensor's output V_{out}.

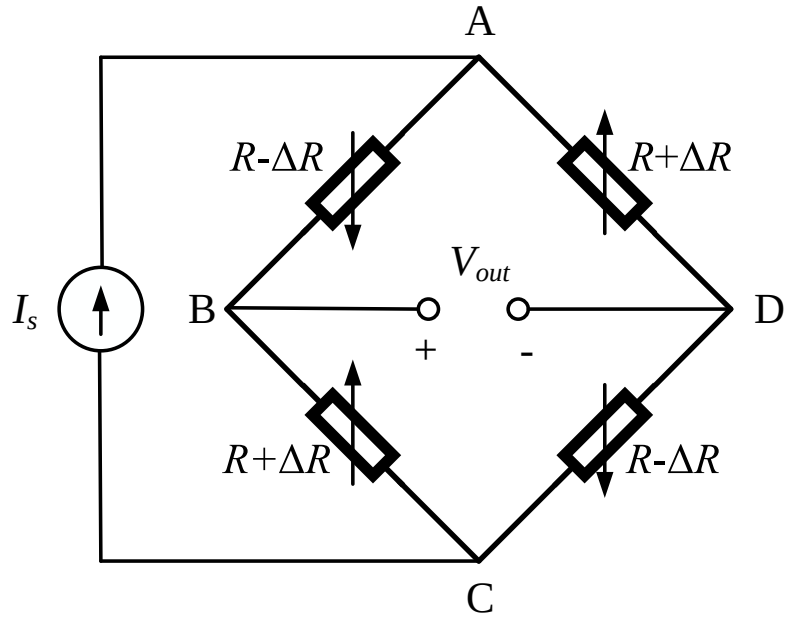


Wheatstone Bridge Driven by a Constant Current



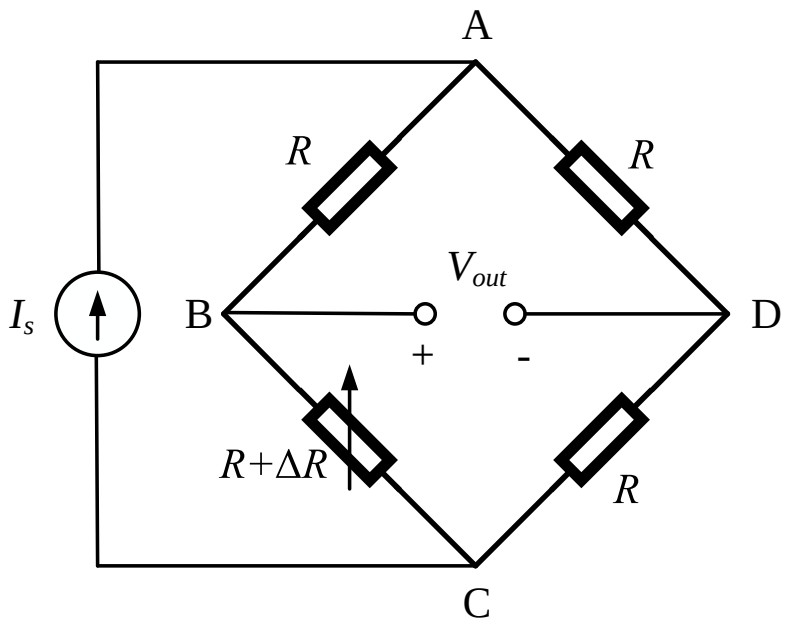


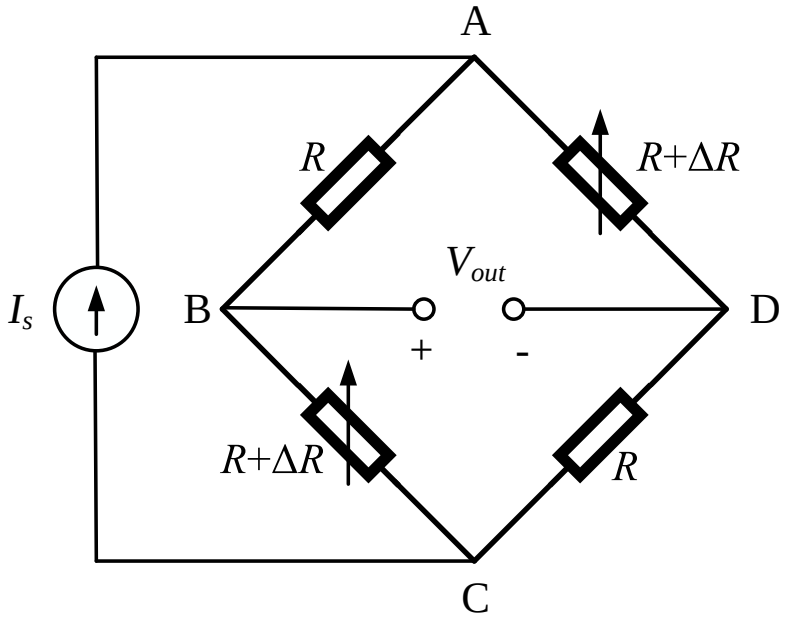




Compare the performance

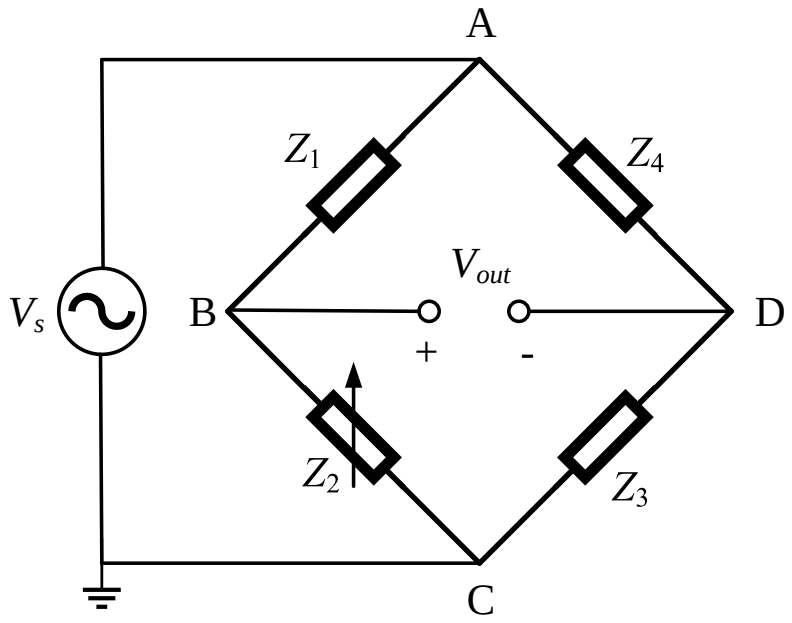
A Wheatstone bridge driven by a constant voltage and a constant current with the configuration of: (1) the single-element variation; (2) the two-element variation (diagonally mounted). Assume that the nominal resistance R is 120Ω, ΔR changes from -100Ω to +100Ω. The constant voltage source is 3VDC, and the constant current source is 25mA.





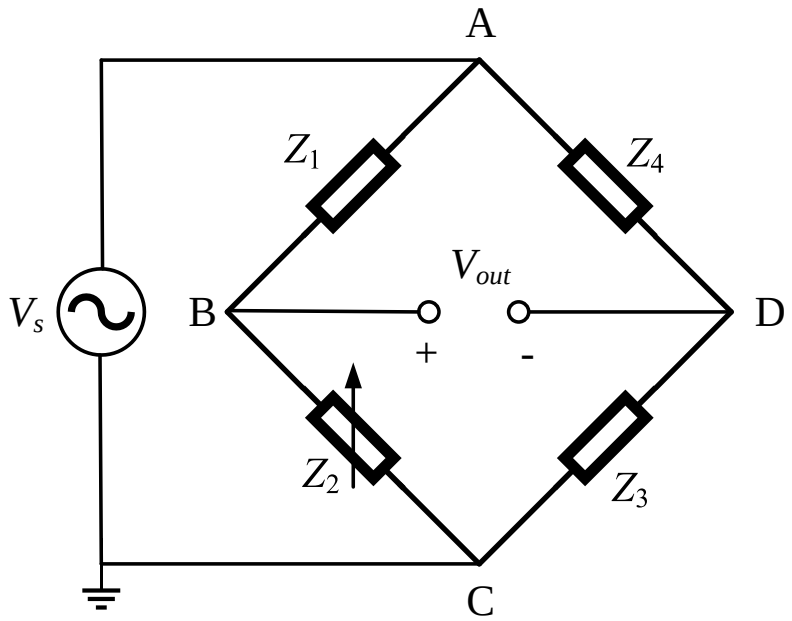
AC bridge-1

An AC bridge has the following impedance: Z_1=100 Ω∠70° (inductor); Z_2 = 300 Ω∠-30° (capacitor); Z_4 = 250 Ω∠0° (resistor). Determine the unknown impedance of Z_3



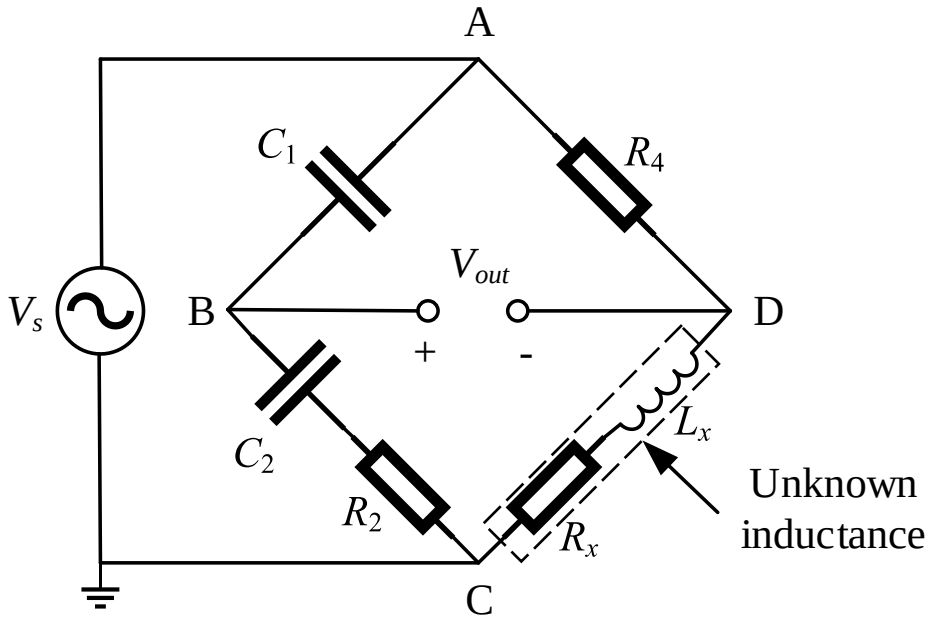
AC bridge-2

The AC bridge is in balance with the following elements: (1) Arm *AB*: $R = 200\ \Omega$ in series with L=15.9mH; (2) Arm *BC*: R=450 Ω; (3) Arm *AD*: $R = 300\Omega$ in series with C= 0.265μF. The AC oscillator frequency is 1 kHz. Find the unknown element in Arm *CD*.



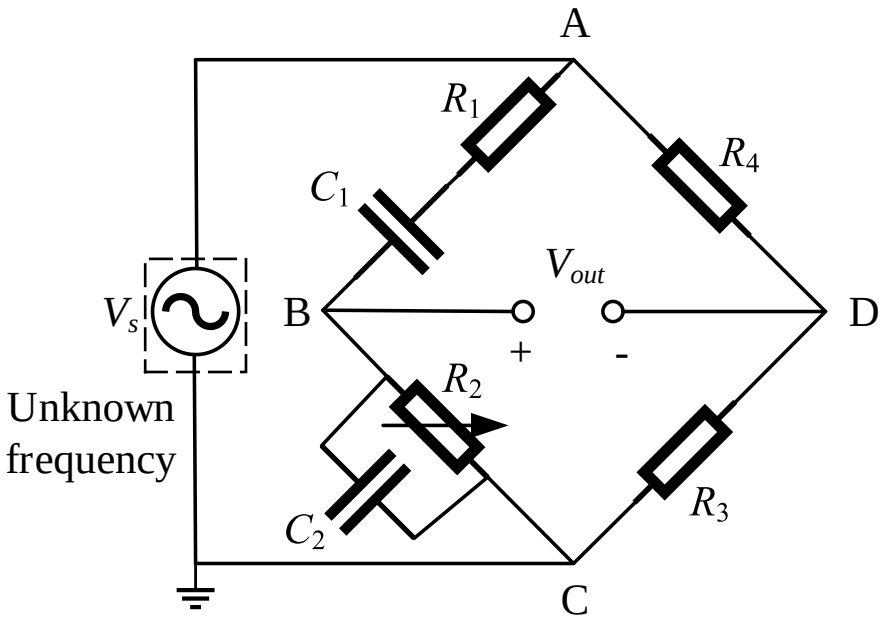
Owen bridge

An Owen bridge reaches a balance point when C_1= 0.8μF, C_2= 0.5μF, R_2=400Ω, and $R4$=450 Ω under a 1 kHz excitation frequency. Find Zx



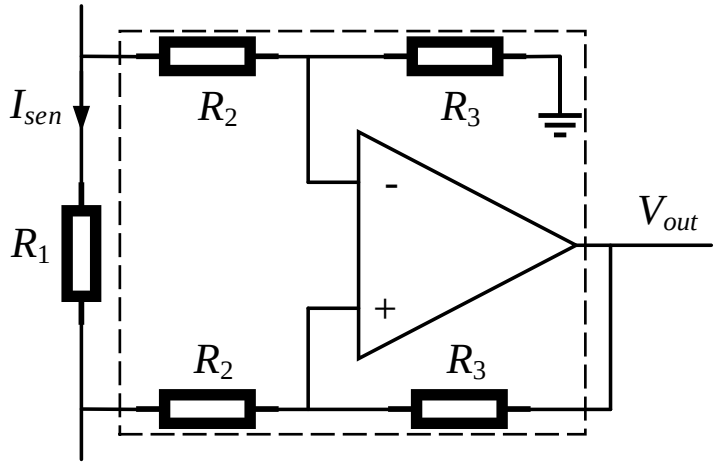
Vien bridge

A Wien bridge is often used to measure frequency of the voltage source using series *RC* in one arm and parallel *RC* in the adjacent arm. Calculate the frequency of the source.

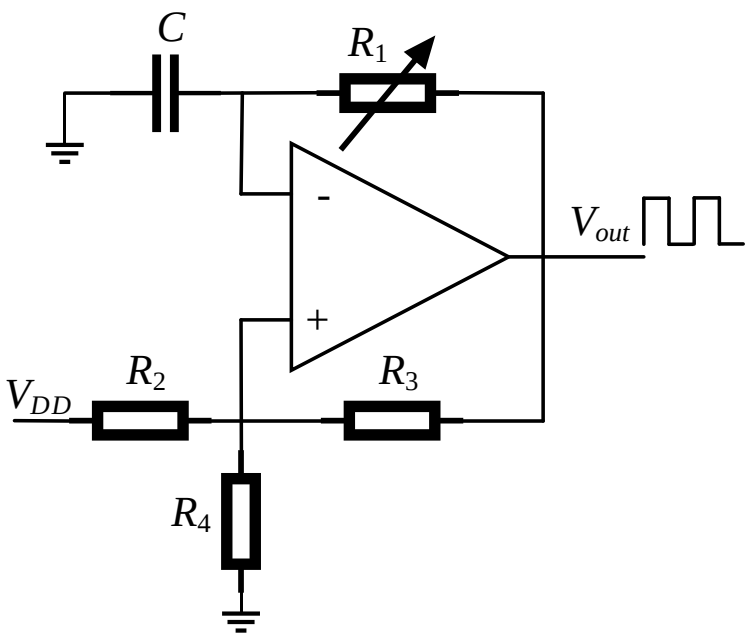


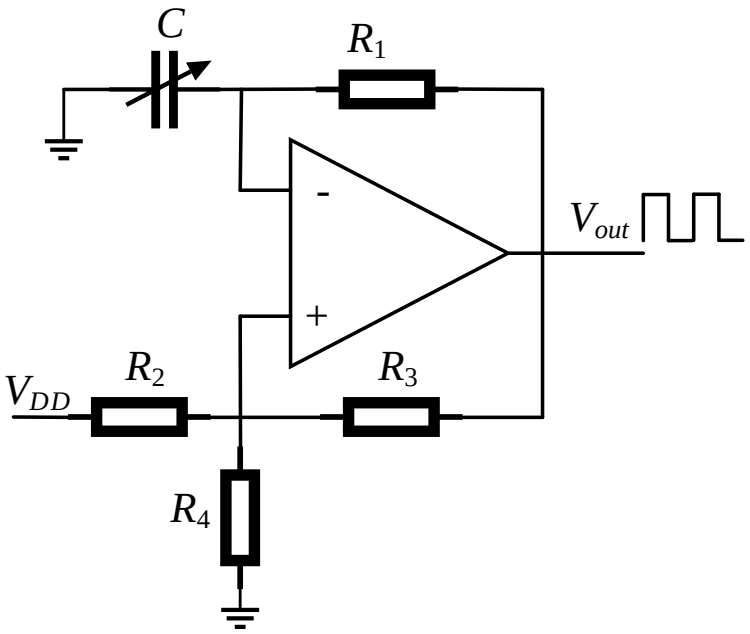
Current output circuits

Current output sensors, such as DC current sensors, certain humidity sensors, and CMOS temperature sensors, output a current instead of a voltage. In this situation, a resistor (e.g., R_1) is often used to convert sensor current output I_{sen} into a voltage output. Then, a differential amplifer amplifies the voltage across the resistor R_1. One problem with this method is that the resistor R_1 loads the sensor output. To reduce this loading, the value of resistance R_1 should be chosen much less than R_2. The output V_{out} expressed is?



Resistance-to-Frequency Conversion: RC Oscillator





Concept question

Which of the following signals can NOT be converted into time by an RC decay circuit?

A. Resistance

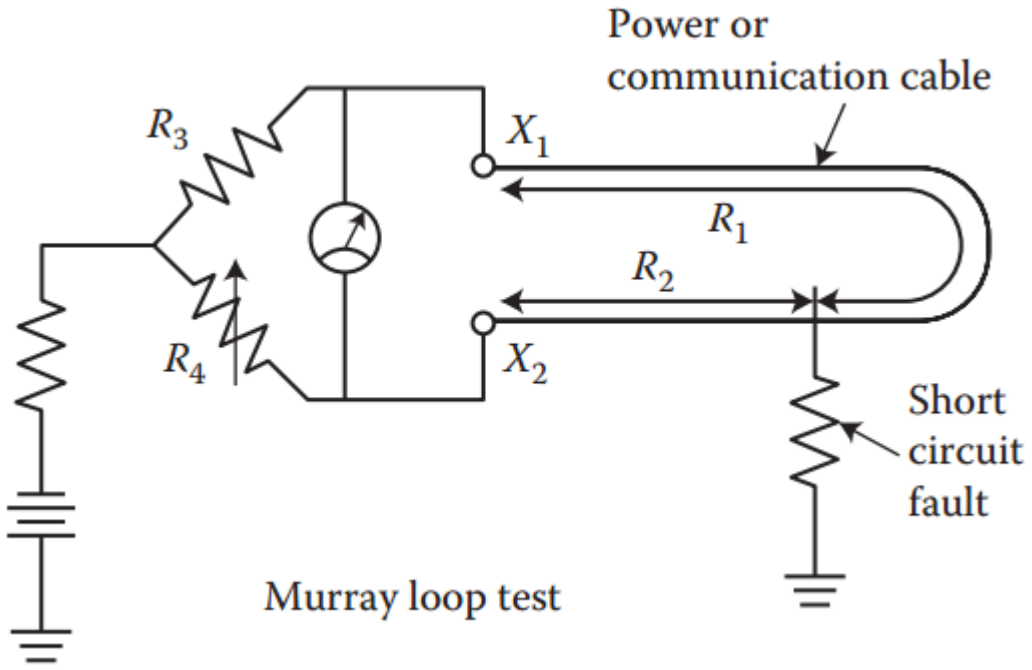
B. Capacitance

C. Voltage

D. Frequency

Question

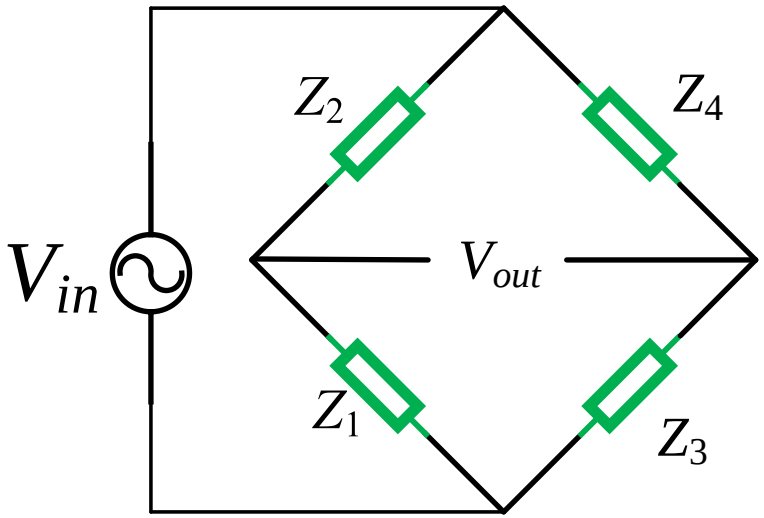
Murray loop test is used to locate ground fault in a telephone system. The total resistance R (= R_1 + R_2) is measured by a Wheatstone bridge, and its value is 255 Ω. The conditions for Murray loop test are: R_3 = 900 Ω and R_4 = 510 Ω. Find the location of the fault in meter, if the length per Ohm is 34.89 m (Hint: a bridge balance condition: $R_1R_4 = R_2R_3$. After finding R_1 and R_2, calculate $R_1 \times 34.89$ for X_1 and $R_2 \times 34.89$ for X_2.)



Murray loop test

Example

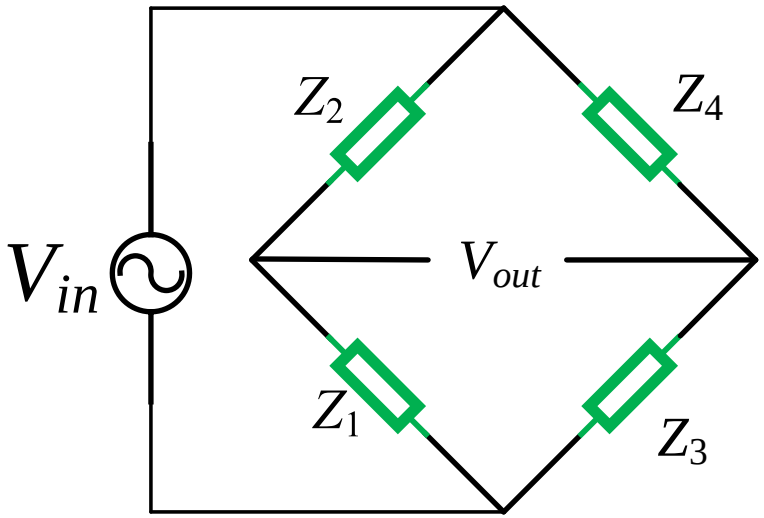
An AC bridge has the following impedance: Z_1 = 100 Ω ∠70° (inductor); Z_2 = 300 Ω ∠ −30° (capacitor); Z_4 = 250 Ω ∠0° (resistor). Determine the unknown impedance of Z_3.



Solution

The AC bridge balancing requires Z_3 to be

$$|Z_3| = \frac{|Z_1||Z_4|}{|Z_2|} = \frac{\mathbf{100 \cdot 250}}{\mathbf{300}} = \mathbf{83.3\Omega}$$



The phase angle balancing requires θ_3 to be

$$\angle\theta_3 = \angle\theta_1 + \angle\theta_4 - \angle\theta_2 = \mathbf{70} + \mathbf{0} - (\mathbf{-30}) = \mathbf{100^\circ}$$

Thus, $Z_3 = 750\ \Omega\ \angle 100^\circ$, which is a inductive element